

DOMINIC L. MARCHESE

Atlanta, Georgia | [Personal Website](#) | [LinkedIn](#)

EDUCATION

Drexel University, School of Economics

B.S. in Economics and Mathematics, magna cum laude.

Philadelphia, PA | 2026

Laurel Ridge Community College

A.S. in Business Administration. Completed concurrently with high school.

GPA: 3.79

Stephens City, VA | 2023

GPA: 3.90

RESEARCH EXPERIENCE

Federal Reserve Bank of Atlanta

Atlanta, GA | 2026 - Present

Research Analyst | Dr. Choongryul Yang | Dr. David Altig

- Support macroeconomic research on monetary and fiscal policy, inflation dynamics, and business-dynamism.
- Conduct statistical and computational analysis for research on inflation expectation formation, diffusion mechanisms, and the transmission of monetary policy to expectations and economic outcomes.
- Contribute to policy-relevant research workflows involving economic data, empirical analysis, and quantitative macroeconomic modeling.
- Contribute to the Career Ladder Identifier and Financial Forecaster (CLIFF) tools with Dr. David Altig.

C. & J. Nyheim Plasma Institute, Drexel University

Camden, NJ | 2025 - 2026

Research Assistant, Computational Physics Laboratory | Dr. Dmitri Vainchtein

- Studied nonlinear wave-particle interactions and energetic-electron transport in Earth's magnetosphere using satellite observations, statistical analysis, and numerical modeling.
- Built a pipeline for approximately 10 TBs of ELFIN satellite data using Python, producing reduced samples for studies of energetic-electron precipitation and magnetospheric dynamics.
- Developed and analyzed large-scale test-particle simulations in Julia to characterize stochastic and non-Fickian transport arising from nonlinear resonant interactions with electromagnetic waves.
- Applied statistical fitting and computational methods to connect spacecraft measurements with analytical and numerical models of space-plasma dynamics.

Drexel University, School of Economics

Philadelphia, PA | 2025 - 2026

Research Apprenticeship | Dr. Christopher Laincz

- Investigated computational endogenous-growth models that track firm age to study aggregate growth slowdowns and barriers to firm expansion.

RESEARCH PROJECTS

Electron Super- and Sub-Diffusion in Velocity Space Due to Nonlinear Resonance with Whistler-Mode Waves 2025 - 2026

- Investigated stochastic transport of energetic electrons under nonlinear resonant wave-particle interactions using theoretical models, empirically fitted inputs, and large-scale numerical simulation.

TEACHING EXPERIENCE

Drexel University

Philadelphia, PA | 2024 - 2025

Tutor, Microeconomics and Macroeconomics

- Tutored more than 15 students in introductory economics and finance, adapting explanations and material coverage to individual students and faculty requests.

ACADEMIC & PROFESSIONAL ACTIVITIES

Federal Reserve College Challenge

2025

- Lead presenter on the Economic Outlook team, with primary responsibility for consumer-health analysis and the acquisition, maintenance, and interpretation of consumer-oriented economic data.

Omicron Delta Epsilon

2025 - Present

Member, international honor society in economics.

Drexel Economics Society

2023 - Present

Member.

RESEARCH INTERESTS

Statistical inference; structural modeling; econometrics; industrial organization; macroeconomics

TECHNICAL SKILLS

Programming and statistical computing: Python, R, Julia, Stata, SQL, LaTeX

Practiced Methods: econometric analysis, statistical modeling, numerical simulation, computational modeling, data visualization